

# SHENGXIAN “NICK” HAO

San Diego, California | nihao@ucsd.edu | (530) -204-7415 | www.linkedin.com/in/nick-hao

## EDUCATION

**University of California, San Diego** – La Jolla, CA, United States

**June 2027**

*M.S. in Bioengineering, Cumulative GPA: 4.0*

**University of California, Davis** – Davis, CA, United States

**June 2025**

*B.S. in Electrical Engineering, Cumulative GPA: 3.42*

## EXPERIENCE

**University of California, San Diego** – La Jolla, CA, United States

**July 2025 – Present**

*Research Assistant Under Professor Francisco Contijoch (August 2025 – Present)*

- Resolved memory allocation issues for k-space data and implemented data binning and buffer management in Siemens MRI sequence programming (C++), and provided detailed logs for team to debug and extend features
- Documented software features and workflow for improved team knowledge sharing and onboarding
- Refactored redundant modules, reduced codebase size by ~8% (measured in lines of code), and improved maintainability

**University of California, Davis** – Davis, CA, United States

**September 2024 – March 2025**

*Research Assistant Under Professor Weijian Yang*

- Participated in advanced research utilizing two-photon excitation technology combined with MEMS components and camera sensors to capture and analyze neuronal activity in mice accurately
- Debugged MEMS-based imaging circuits using oscilloscopes and multimeters, ensuring stable data acquisition
- Enhanced understanding of photonic applications through hands-on printed circuit board design
- Redesigned an open-source Miniscope PCB into a custom sandwich architecture using flex cables and standoffs, achieved ~38% reduction in footprint while maintaining full functionality

## PROJECTS

**Physics-Constrained GNN for Liver Hemodynamic (Course Project)** – La Jolla, CA, United States

**January 2026 – Present**

- Engineered a Physics-Constrained Graph Neural Network (PC-GNN) surrogate model using the MeshGraphNet architecture and NVIDIA PhysicsNeMo to simulate 1D hemodynamics in complex liver vascular networks.
- Embedded fundamental physical laws into the training process by implementing a custom multi-objective loss function
- Accelerated hemodynamic inference by 5000x compared to traditional CFD simulations

**Neuromorphic Vision Silicon Chip (Course Project)** - La Jolla, CA, United States

**September 2025 – December 2026**

- Designed a transistor-level silicon retina pixel in SKY130 CMOS process implementing local light-offset cancellation
- Reduced reset noise by >90% through correlated double sampling

**Gesture-Controlled Robotic System (Course Project)** - Davis, CA, United States

**September 2024 – March 2025**

- Developed hardware for robotic arm control. Designed custom PCB; programmed firmware in C
- Integrated real-time motion simulation in MATLAB with limit detection in 6 axes

## LEADERSHIP

**IEEE UC Davis Student Branch** – Davis, CA, United States

**September 2023 – June 2025**

*Vice President*

- Coordinated with industry leaders, including HPE and Texas Instruments, to host information sessions while assisting the Chair in managing branch activities, and supporting project teams in securing funding and resources.
- Proposed the Smart Agriculture Project for 28 students across four areas: Smart Irrigation System, Power Supply System, Computer Vision, and Smart Automatic Vehicle. Coordinated a student event to encourage engagement, assisted students in gathering project information, and provided guidance to achieve project goals while managing multiple responsibilities.

## SKILLS & INTERESTS

- Coding Language: C, C++, Python(PyTorch), Verilog, Matlab, LaTeX, Git
- Industry Software: OrCAD, Intel Quartus, PSoC Creator, KiCAD, Altium, LTspice, Arduino IDE, Keil Uvision 5, STM32CUBE, Cadence Virtuoso, Silvaco Deckbuild, ModelSim, Siemens IDEA
- Relevant Coursework: Electronic Circuits, Digital Systems, VLSI Design, Integrated Circuit Fabrication, Signals & Control Systems, Embedded System Design, Applied Probability, Device Physics, Machine Learning for Medical Imaging, Biology